



A Proposed Mathematical Model for the Cyclic Relationship between Estradiol and Endometriosis Lesions

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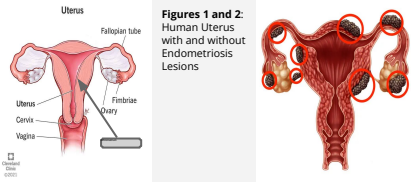
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Introduction



Figures 1 and 2:
Human Uterus
with and without
Endometriosis
Lesions

• **Endometriosis** is a biological condition in which **endometrial tissue** continuously **grows outside the endometrium**, forming endometriosis lesions (2).

• Estradiol, E2, and progesterone, P4, are hormones that support the uterus throughout the menstrual cycle. **When the ratio of E2 to P4 is too high, E2 over-communicates with the body, resulting in uncontrollable endometrial growth.**

• **Endometriosis lesions produce estradiol**, contributing to a cycle of continuous estradiol and progesterone imbalance.

• **Common treatment is exogenous estradiol suppression.**

Model Development

• The proposed model expanded on **Margolskee et al.'s model of the baseline menstrual cycle** and **Arbeláez-Gómez et al.'s model of endometrium growth** (3).

• The model proposed mathematically **mimicked endometriosis lesion volume growth over time** and **the relationship between lesion volume and estradiol** (4).

• **Lesion growth was stimulated by feeding estradiol and progesterone levels**, from Margolskee et al.'s model, **into the proposed model** (3).

• **A percentage of lesion volume** from Arbeláez-Gómez et al.'s model, V_{TL} , **was added to the E2 equation, increasing estradiol levels** (5).

• Elevated E2 promoted the continuous growth of the lesions, **creating a cyclic relationship between E2 and the endometriosis lesions.**

Figure 4: Differential Equations used to Model Endometriosis Lesion Growth

$$\begin{aligned} 1 \frac{dM_{X_{proL}}}{dt} &= \mu_{proL} M_{X_{proL}} - \mu_{secL} M_{X_{proL}} \\ 2 \frac{dM_{X_{secL}}}{dt} &= Y_{X_{secL}} \mu_{secL} M_{X_{proL}} - \dot{m}_6 w_{X_{secL}} \\ 3 \frac{dM_{X_{pL}}}{dt} &= q_{pL} M_{X_{secL}} - \dot{m}_6 w_{pL} \end{aligned}$$

$$V_{TL} = \frac{M_{X_{proL}}}{\rho_{X_{proL}}} + \frac{M_{X_{secL}}}{\rho_{X_{secL}}} + \frac{M_{X_{pL}}}{\rho_{X_{pL}}}$$

Figure 5: Equation for the Overall Estradiol Levels after an Endometriosis Lesion is Introduced (V_{TL})

$$E2 = e_0 + e_1 GrF + e_2 DomF + e_3 Lut4 + e_4 V_{TL}$$

Results

Figure 6: Endometriosis Lesion Growth, as Excess Estradiol is Introduced into the System

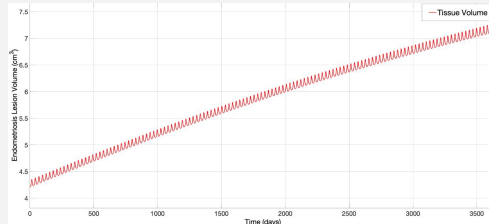
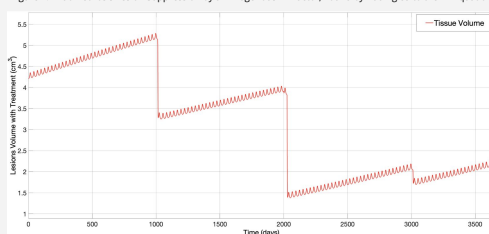


Figure 7: Endometriosis Lesion Suppression by an Exogenous E2 Factor, Found by Adding e_5 to the E2 Equation



Discussion

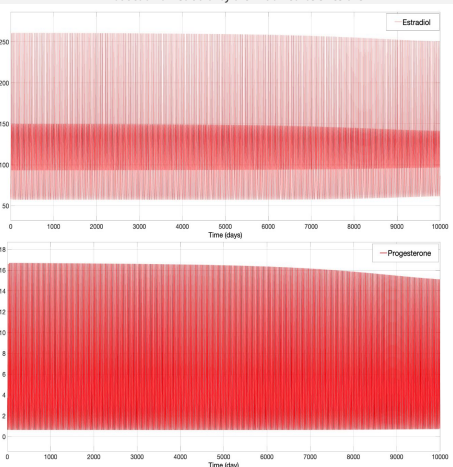
• When endometriosis lesion volume was factored into the equation of E2 (5), **estradiol levels in the body increased, causing the lesions to grow even larger** (6).

• **Estradiol levels did not change as expected**, even though the E2 equation was altered based on the endometriosis lesion volume (8).

• **Estradiol decreased faster than progesterone** did (8), indicating E2 dominance and P4 resistance.

• When an **exogenous estradiol suppressor** was introduced to the E2 equation, **endometriosis lesion volume decreased** (7).

Figure 8: Changes in Estradiol and Progesterone Levels Over 10,000 Days, Caused by the Production of Estradiol by the Endometriosis Lesions



Conclusion

• Two causes of potential error were **non biologically accurate parameters** and a **non biologically accurate depiction of how tissue from the endometrium plants itself outside of the endometrial cavity.**

• Going forward, how a **predictive model could be used to categorize endometriosis lesion size**, how other hormones, such as GnRH, influence the model, and the effects of treatment on the model will be explored.

• In creating this model, **endocrinologists and medical professionals can further explore the apparent relationship between estradiol and long-term endometriosis lesion growth.**

